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William Nordhaus and the Economics of Climate Change

The 2018 Nobel Memorial Prize in Economics was awarded to two American Economists, Paul Romer and William Nordhaus, for their work on economic growth theory and the economic impact of climate

change, respectively. This article will attempt to scratch the surface of William Nordhaus' breakthrough contribution to economic modelling of climate changes, or in the words of the prize committee, "for integrating climate change into long-run macroeconomic analysis."

One primary area that Professor Nordhaus pioneered in is known as "Green Accounting," which attempts to outline how environmental degradation can be measured against economic growth, starting in 1972. In the 1990s, Professor Nordhaus published the now groundbreaking Dynamic Integrated Climate-Economy (DICE) model, which is the first ever major attempt to develop a method of estimating the economic costs of climate change.

In the words of Nordhaus and his collaborators, it "integrate[s] in an end-to-end fashion the economics, carbon cycle, climate science, and impacts in a highly aggregated model that allow[s] a weighing of the costs and benefits of taking steps to slow greenhouse warming." Since the model allows economists and policy makers to simulate how the environment and the economy will evolve together under various different assumption, it has now become one of the main analytical tool to measure and quantify the social and economic costs of carbon.

Professor Nordhaus is also an advocate of government controls over fossil fuels, which is a global public good. His main argument is that unless governments include the negative external effects by raising the price of fossil fuels, firms would essentially become "free-riders" to take the benefit of these natural resources without paying their environmental costs.

One of Nordhaus' solutions to global climate change is the notion of a "global climate club," where a great number of countries would set a world carbon price together through a carbon tax or a cap-and-trade system, which prevents the over-consumption of the environment. Countries refusing to join the club will be punished with tariffs and

other sanctions, which will propel them to eventually join the club once the cost of these punishments becomes large enough.

Nordhaus' wide research interests have also lead to numerous papers in other emerging fields of economics. In particular, his work on the political business cycle theory is also fascinating, which explores how politicians boost their chances of reelection by manipulating the economy using fiscal and monetary policies.

Coincidentally, in the same week that the 2018 awards were honoured, a report by UN has warned the public of catastrophic climate effects if we do not limit carbon emission in the next 12 years. Of course, although groundbreaking, Professor Nordhaus' work on climate change is definitely not conclusive. Nonetheless, it serves as a great foundation for future generations of economists to continue to raise public awareness on the topic while offering feasible solutions to this urgent and pressing issue.

About Professor Nordhaus:

After receiving a BA and MA from Yale, Professor Nordhaus obtained his PhD in MIT and has served as a Yale faculty since 1967. As the Sterling Professor of Economics at Yale University, Professor Nordhaus has also served various roles as government advisors and chairman of the BoD of the Boston Federal Reserve Bank. In addition to his extraordinary academic work, Nordhaus is also extremely well-known as the coauthor of the monumental introductory textbook Economics, which were originally authored by the 1969 Nobel Laureate and one of the greatest economists of all time, Paul Samuelson.

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